Firm Dynamics

Hopenhayn, H. 1992. "Entry, Exit, and Firm Dynamics in Long Run Equilibrium", *Econometrica*, 60, p. 1127-50. (partial equilibrium, discrete choice problem to derive firm size distributions).

Ericson, R. and A. Pakes. 1995. "Markov-Perfect Industry Dynamics: A Framework for Empirical Work", *Review of Economic Studies*, 62, p. 53-82 (see especially Section IV. An Example. Since this paper considers an environment with a homogeneous good, it is closer to Hopenhayn than the "seminal" paper Pakes, A. and P. McGuire (1994), "Computing Markov-Perfect Nash equilibria: Numerical implications of a dynamic differentiated-product model", RAND, 25, 555-89. See also http://www.economics.harvard.edu/faculty/pakes/program.html).

Gowrisankaran, G. and T. Holmes (2004) "Mergers and the Evolution of Industry Concentration: Results from the Dominant-Firm Model," *RAND Journal of Economics*, 35(3), p. 561-582. This paper considers a Stackelberg game which mixes the two; there is a monopolist that moves first and a competitive fringe that follows.

1 Environment

- Population: Industry composed of different numbers of firms conditional on market structure considered - a continuum of firms with mass not necessarily 1 in the competitive market (Hopenhayn) and a finite number for the monopolistic and duopoly case (Ericson and Pakes).
- Each firm produces a homogeneous product.
- Aggregate Demand for the final good given by the *inverse demand* function $P(Q_t)$. <u>Assumption A.1a</u>: P is continuous, decreasing, and $\lim_{Q\to\infty} P(Q) = 0$.
  - For example, let $P = \max\{D - Q, 0\}$ where $D = 5$.
- <u>Assumption A.1.b</u>: The input price $W(N_t)$,where N_t is total industry labor demand, is assumed to satisfy properties: W is continuous, nondecreasing and strictly bounded above zero.
  - For example, let $w = 1$.
- Technology: output of a given firm is $q_t = f(\varphi_t, n_t)$ where $\varphi_t \in [0, 1]$ is a productivity shock which follows a Markov process, independent across firms with conditional distribution $F(\varphi_t|\varphi_{t-1})$.
  - For example, let $f(\varphi, n) = \varphi n^\alpha$ where $\alpha = \frac{1}{2}$. It is important that the production function exhibits decreasing returns to scale.

- A fixed cost $w_t c_f$ must be paid every period by an operating firm (for exit to take place). Here we will assume that the cost is associated with management. We will assume that you do not have to pay the cost the period you exit.
  - For example, we will take $c_f = 1.9$
- Profit function given by $\pi(\varphi_t; P, W) = max_n Pq_t - Wn_t - Wc_f$.
- Firms discount profits at rate $0 < \beta < 1$.
  - For example, $\beta = .96$ (time period of 1 year at 4%).
- Assumption A.2.(a) q and n are single valued, increasing in φ, and continuous; (b) π is continuous and strictly increasing in φ; (c) $\lim_{Q \to 0} \pi(\varepsilon; P(Q), w) > 0$.
- Assumption A.3(a) F is continuous in φ_{t-1} and φ_t, (b) F is strictly decreasing in φ_{t-1}. (basically just persistence).
- Assumption A.4 (Recurrence) For any $\varepsilon > 0$, there exists an integer k such that $F^k(\cdot|\varphi)$ gives the distribution of φ_{t+k} given $\varphi_{t-1} = \varphi$ (a mixing condition).
  - For example, suppose there are two states. Let $\varphi \in \{\varphi_L, \varphi_H\}$ with $1 = \varphi_H > \varphi_L = .6 > 0$. $F(\varphi' = \varphi_{j \neq i} | \varphi = \varphi_i) = 1 - \theta < \frac{1}{2}$, $i, j \in \{H, L\}$. Thus, the process displays persistence since $\theta > \frac{1}{2}$. We take $\theta = 0.9$.
- A.2 and A.3 imply expected discounted profits are increasing in φ. Furthermore, if your profits are high (low) today, they are expected to stay high (low) in the near future. This is important for the exit decision.
- Each period, after φ_t is realized, incumbent firms may exit. An exiting firm receives 0 present value.
- In equilibrium, firms exit whenever their state φ_{t-1} falls below a reservation level x_{t-1}. Given this cutoff point, A.4 implies the life span of a firm is almost surely finite (i.e. there is a steady stream of exit).
- An entering firm must pay a fixed cost $w_t c_e$. Then it draws a productivity shock φ_t from initial ergodic density function η.
  - For example, let $\Pr\{\varphi = \varphi_H\} = \eta = \frac{1}{2}$ and $c_e = 3.0$.
- Assumption A.5 η has a continuous distribution function G.
- Summary of timing: Incumbent (same as Ericson and Pakes)

1. Shock is realized φ_t.

2. Incumbent firm may exit to avoid c_f.

3. If incumbent stays, pays fixed cost c_f and chooses labor demand $n = g(\varphi_t; P, W)$ to earn profit $\pi_(\varphi_t; P, W)$

- Entering firm must pay fixed cost c_e one period in advance (kind of like one period time to build) after stage 3 (**different**). For simplicity, we will assume the following cost structure. In the case of Hopenhayn, this is just a number. In the case of Ericson-Pakes,

$$c_e = \begin{cases} c & \text{if number of incumbent firms} \leq NF \\ \infty & \text{if number of incumbent firms} > NF \end{cases}$$

- Summary of timing: Potential Entrant

1. Decide whether to pay fixed cost wc_e in period $t-1$

2. Receive this period's shock φ_t from $\eta(\varphi_t)$

3. Same as 2-3 above[1]

- Let the measure $\mu_t(A)$ summarize the number of incumbent and entrant firms that have $\varphi_t \in A \subset S$. Due to entry and exit, total mass of firms $\mu_t(S)$ is not necessarily 1. This can be thought of as being evaluated before the exit decision is made (stage 2) but after the realization of φ.

- Notice that under the timing above, even if firms with $\varphi_t \leq x_t$ are exiting, the distribution $\mu_t(\varphi_t)$ includes new entrants who may draw low φ_t from the distribution function $G(\varphi_t)$ and will immediately exit.

- Aggregate output supply and input demand given by

$$Q_t^s(\mu_t; p, w) = \int q_t(\varphi_t; p, w)\mu_t(d\varphi_t) \tag{1}$$

$$N_t^d(\mu_t; p, w) = \int n_t(\varphi_t; p, w)\mu_t(d\varphi_t) \tag{2}$$

2 Competitive Equilibrium (Hopenhayn)

- We assume a continuum of firms ($NF = \infty$), with mass not necessarily equal to 1. Firms behave competitively, taking prices in the output (p_t) and labor input (w_t) markets as given.

- Given that all the uncertainty is idiosyncratic and there is a continuum of firms, we will be considering stationary equilibria where $\{p_t, w_t\} = \{p, w\}$. By the Law of Large Numbers, there is no aggregate uncertainty in this case.

[1] Note that under this timing an entrant may immediately exit (which is actually consistent with (**??**) below.

- The problem of an incumbent firm at stage 2 in the above timing is really just a decision on exit (note that the static labor demand problem has already been solved implicitly in the profit function):

$$v(\varphi; p, w) = \max\{0, \pi(\varphi; p, w) + \beta E[v(\varphi'; p, w)|\varphi]\} \tag{3}$$

where $E[g(\varphi')|\varphi] = \int_{\varphi' \in [0,1]} g(\varphi') F(d\varphi'|\varphi)$.[2]

- The solution to problem (3) is a cutoff rule. Let

$$x(p, w) = \inf\{\varphi \in S : \pi(\varphi; p, w) + \beta E[v(\varphi'; p, w)|\varphi] \geq 0\} \tag{4}$$

Thus any firm with $\varphi < x$ will exit.

- Prop. 1. v is a bounded, continuous, and strictly increasing function of φ. Note: present discounted profits are increasing in productivity follows from properties of π and F.

- Since all firms are ex-ante identical, entry occurs provided

$$\int v(\varphi; p, w)\eta(d\varphi) \geq wc_e.$$

- Let M_t denote the mass of entrants in period t. An equilibrium with free entry requires

$$\int v(\varphi; p, w)\eta(d\varphi) \leq wc_e, \tag{5}$$

with equality if $M_t > 0$ (in which case we are free to "choose" any measure M_t we want).

- The entry and exit rules imply an evolution for the state of the industry

$$\mu_{t+1}([0, \overline{\varphi}']) = \int_{\varphi' \in [0, \overline{\varphi}']} \int_{\varphi \geq x} dF(\varphi'|\varphi)\mu_t(d\varphi)d\varphi' + M_{t+1}G(\overline{\varphi}') \tag{6}$$

Note that the first term considers only those firms which do not exit.

- Could write this in terms of Transition function

$$T(\varphi, A) = \begin{cases} \int_{s' \in A} dF(s'|\varphi)ds' & \text{if } \varphi \geq x \\ 0 & \text{otherwise} \end{cases}$$

so that (6) given by

$$\mu_{t+1} = T\mu_t + M_{t+1}\eta \tag{7}$$

- *Importantly*, it is simple to see from (7) that μ_{t+1} is homogeneous of degree 1 in μ_t and M_{t+1}.

[2] In Hopenhayn's notation, $F(d\varphi'|\varphi)$ stands for the densitiy function for next period's shock φ' conditional on φ.

Definition 1. *A* ***stationary competitive industry equilibrium*** *is a list* $\{p^*, w^*, Q^*, N^*, M^*, x^*, \mu^*\}$ *such that*

1. $p^* = D(Q^*)$, $w^* = W(N^*), Q^* = Q^s(\mu^*, p^*, w^*)$, $N^* = N^d(\mu^*, p^*, w^*)$ (markets clear)
2. x^* satisfies (4) (optimal exit)
3. (5) satisfied at p^* and w^* with equality if $M^* > 0$ (free entry)
4. $\mu_t = \mu^*$ solves (7)

- To prove existence, find a fixed point of $\mu \to (p(\mu), w(\mu)) \to (v, g) \to \mu$.
- Theorem 2. A SS equilibrium exists (there are many possible equilibria). e.g. no trade is always an equilibrium.
- Theorem 3. A SS equilibrium with entry/exit exists iff entry costs are sufficiently low.
- Theorem 4. Conditions for uniqueness include production functions of the form $f(\varphi, n) = \varphi n^\alpha$, $\alpha \in (0, 1)$. [3]
- An important step in Theorem 2 is existence of a unique invariant measure μ^* which follows from the recurrence or mixing condition of A.4.

3 Competitive Example (NF=∞)

The environment is an example of the above and will be similar to that of Hopenhayn. We take the following parameterization:

Parameter	Value
Inverse Demand	$P(Q) = \max\{D - Q, 0\}$, $D = 5$
Wage	$w = 1$
Production Function	$f(\varphi, n) = \varphi n^\alpha, \alpha = \frac{1}{2}$
Productivity Grid	$\varphi \in \{\varphi_L, \varphi_H\} = \{0.6, 1\}$
Productivity Persistence	$Pr(\varphi_{t+1} = \varphi \vert \varphi_t = \varphi) = \theta = 0.9$
Entrant distribution	$Pr(\varphi = \varphi_H) = \eta = \frac{1}{2}$
Discount Factor	$\beta = 0.96$
Operation Cost	$c_f = 1.9$
Entry Cost	$c_e = 3$

Table 1:

[3] This is condition U.2 on p. 1137.

- The inverse demand function is taken exogenously but can be viewed as the result of a representative household solving the following static optimization problem with a normalization of wages ($W = 1$):

$$\max_{\{C,N\}} \quad DC - \frac{C^2}{2} - N$$
$$\text{s.t.} \quad PC = WN + \Pi$$

- where $\Pi = \int \pi(\varphi, P)\mu(d\varphi)$ is the aggregate firm profits.
- firm's static optimization problem:

$$\left[D - \int q(\varphi; P(\mu))\mu(d\varphi)\right] \alpha\varphi_i n^{\alpha-1} = 1 \Longleftrightarrow n_i^* = (P\alpha\varphi_i)^{1/(1-\alpha)}, i \in \{H, L\} \tag{8}$$

  In this specific case, $n_i^* = \left(\frac{P\varphi_i}{2}\right)^2$.

- Thus

$$\begin{aligned} \pi(\varphi_i, P, 1) &= P\varphi_i \left(n_i^*\right)^\alpha - n_i^* - c_f \\ &= (P\alpha\varphi_i)^{1/(1-\alpha)} \left[\frac{1}{\alpha} - 1\right] - c_f \end{aligned}$$

  In this specific case, $\pi_i^* = \left(\frac{P\varphi_i}{2}\right)^2 - c_f$.

- The two steady state value functions indexed by φ_i are

$$\begin{aligned} v(\varphi_i; P) &= \max\Big\{0, \pi(\varphi_i, P) + \beta\big[\theta v(\varphi_i; P) + (1-\theta)v(\varphi_{j\neq i}; P)\big]\Big\} \\ &= \max_{x\in\{0,1\}} (1-x)\Big\{\pi(\varphi_i, P) + \beta\big[\theta v(\varphi_i; P) + (1-\theta)v(\varphi_{j\neq i}; P)\big]\Big\} \end{aligned}$$

  where the decision rule $x(\varphi_i; P) = 1$ denotes exit.

- Specifically, $x = 1$ if $\pi(\varphi_i, P) + \beta\big[\theta v(\varphi_i; P) + (1-\theta)v(\varphi_{j\neq i}; P)\big] < 0$.
- Here we will "construct" an equilibrium where if a firm receives a low shock, it will exit (i.e. $\pi(\varphi_L, P) + \beta\big[(1-\theta)v(\varphi_H; P)\big] < 0$) but if it receives a high shock, it will not exit (i.e. $\pi(\varphi_H, P) + \beta\big[\theta v(\varphi_H; P) + (1-\theta)v(\varphi_L; P)\big] > 0$).
- Conjecture this exit strategy is optimal and then check under what conditions it is optimal. Under the construction,

$$v(\varphi_L; P) = 0 \tag{9}$$
$$v(\varphi_H; P) = \frac{\pi(\varphi_H, P)}{1 - \beta\theta} \tag{10}$$

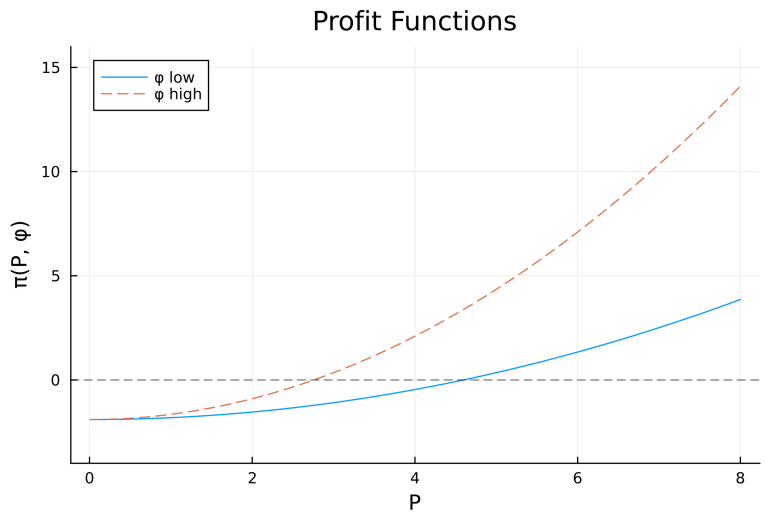

Figure 1: Static Profit Functions

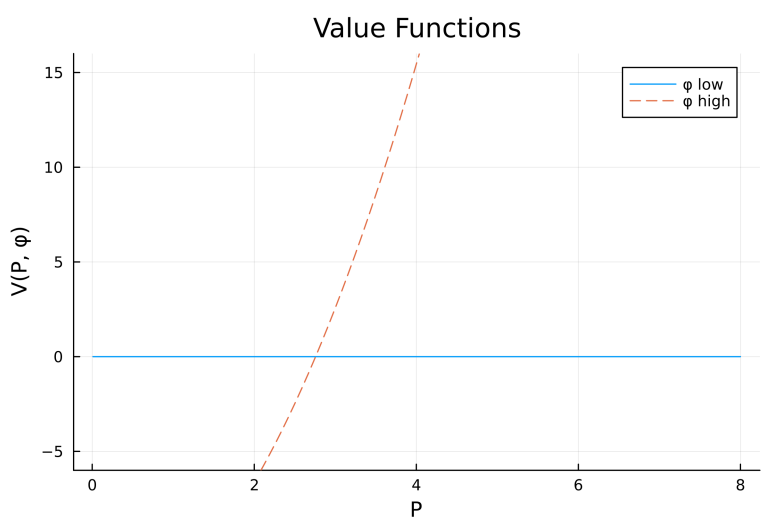

Figure 2: Value Functions

- Since we want positive entry, we must satisfy $\eta \frac{\pi(\varphi_H;P)}{1-\beta\theta} = c_e$. Again, under

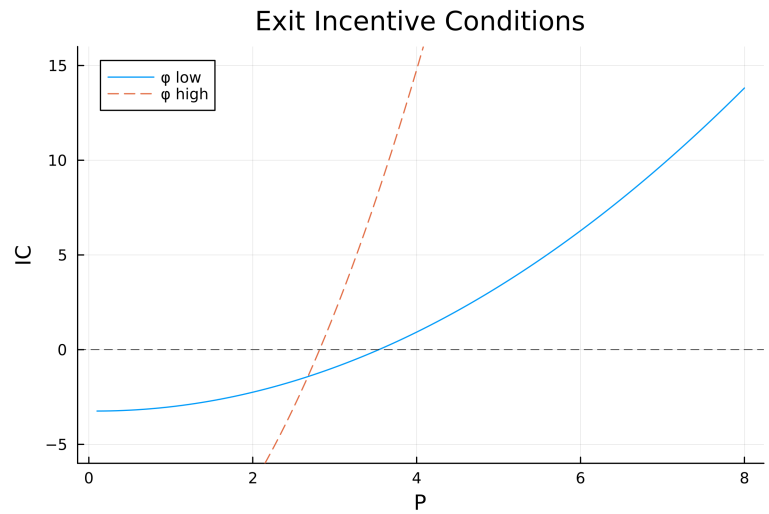

Figure 3: Exit incentives

our conjecture, we have

$$\begin{aligned} c_e &= \eta \frac{(P\alpha\varphi_H)^{1/(1-\alpha)}\left[\frac{1}{\alpha}-1\right]-c_f}{1-\beta\theta} \\ \Rightarrow P &= \frac{\left\{\left[c_e\left(\frac{1-\beta\theta}{\eta}\right)+c_f\right]\left(\frac{\alpha}{1-\alpha}\right)\right\}^{(1-\alpha)}}{\alpha\varphi_H}. \end{aligned}$$

- Now we need to pin down the law of motion for the distribution (which will give us Q) and ultimately M.[4]

$$\begin{aligned} \mu_{t+1}(\varphi_H) &= \mu_t(\varphi_H)\theta + M_t\eta \qquad (11) \\ \mu_{t+1}(\varphi_L) &= \mu_t(\varphi_H)(1-\theta) + (1-x(\varphi_L;P))\mu_t(\varphi_L) + M_t(1-\eta) \end{aligned}$$

  Under the conjectured equilibrium $x(\varphi_L;P)=1$.

- From (11), we can solve for the invariant distribution

$$\begin{aligned} \mu(\varphi_H) &= \frac{\eta M}{(1-\theta)} \\ \mu(\varphi_L) &= \eta M + M(1-\eta) = M \end{aligned}$$

- Output given by

$$\begin{aligned} Q(M) &= \mu(\varphi_H)\varphi_H\left(n_H^*\right)^\alpha \\ &= \frac{M\eta}{1-\theta}\varphi_H\left(n_H^*\right)^\alpha \end{aligned}$$

[4]Here I write the distribution as a pdf rather than a cdf.

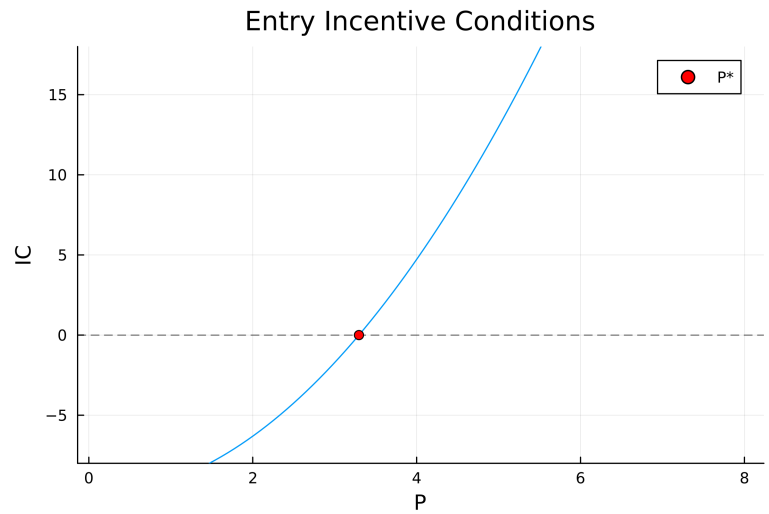

Figure 4: Entry Incentives

That is there are some incumbent firms and new entrants who produce in a given period t after receiving the high shock (if they receive a bad shock in t they exit and they don't produce). Now we can pin down M when we combine this equation with the demand function and the value of P we derived above.

- We need to check that a one period deviation is not profitable, that is, at the equilibrium price, the firms with the low shock prefer to exit (i.e. $\pi(\varphi_L; P) + \beta\big[\theta v(\varphi_L; P) + (1-\theta)v(\varphi_H; P)\big] < 0$). At the above parameter values, this condition holds.

- The next table provides the equilibrium values for this example. Note that aggregate output and prices are state invariant (which will be very different in the next examples).

Competitive Example
$P = 3.296$, $Q = 1.704$, $M = 0.301$
$n(0.6) = 0.994$, $\pi(0.6) = -0.922, v(0.6) = 0$ (off-the-equilibrium path)
$n(1.0) = 2.716$, $\pi(1.0) = 0.816, v(1.0) = 6.0$

4 Monopoly Example (NF=1)

In this case, there is "trivial" aggregate uncertainty due to the idiosyncratic shocks the monopolist receives and the monopolist recognizes that its quantity choice will affect the price it faces.

- firm's static optimization problem:

$$n_i^m = \arg\max_n \left[D - \varphi_i n^\alpha\right] \varphi_i n^\alpha - n - c_f \tag{12}$$

The FOC is

$$P\alpha\varphi_i \left(n^m\right)^{\alpha-1} - \varphi_i^2 \alpha (n^m)^{2\alpha-1} = 1 \tag{13}$$

Using this equation we can solve for the optimal labor choice n_i^m.

- Notice that for a given price, the marginal benefit of adding labor for a monopolist (which is just the left hand side of (13)) is lower than the marginal benefit for a competitive firm (8)) - compare $-\varphi_i^2\alpha(n^m)^{2\alpha-1}$ to 0 - while the marginal cost (which is just the right hand side of the equations) is identical. This is the key factor why the monopoly quantity will be lower than that with perfect competition. Of course, the prices will differ too so this is just the intuition.

- Thus

$$\pi^m(\varphi_i) = \left[D - \varphi_i (n_i^m)^\alpha\right] \varphi_i (n_i^m)^\alpha - n_i^m - c_f$$

- The two steady state value functions indexed by φ_i are

$$\begin{aligned} v^m(\varphi_i) &= \max\Big\{0, \pi^m(\varphi_i) + \beta\big[\theta v^m(\varphi_i) + (1-\theta) v^m(\varphi_{j\neq i})\big]\Big\} \\ &= \max_{x\in\{0,1\}} (1-x)\Big\{\pi^m(\varphi_i) + \beta\big[\theta v^m(\varphi_i) + (1-\theta) v^m(\varphi_{j\neq i})\big]\Big\} \end{aligned}$$

- The monopolist will choose $x(\varphi_i) = 1$ if $\pi^m(\varphi_i) + \beta\big[\theta v^m(\varphi_i) + (1-\theta) v^m(\varphi_{j\neq i})\big] < 0$.

- Here we will "construct" an equilibrium where the monopolist always stays in the market for the same parameter values as the competitive case. This is possible because the monopoly profits are uniformly higher than with perfect competition (where firms with φ_L exit).

  Under the construction,

$$\begin{aligned} v^m(\varphi_L) &= \pi^m(\varphi_L) + \beta\Big[\theta v^m(\varphi_L) + (1-\theta) v^m(\varphi_H)\Big] \\ v^m(\varphi_H) &= \pi^m(\varphi_H) + \beta\Big[\theta v^m(\varphi_H) + (1-\theta) v^m(\varphi_L)\Big] \end{aligned}$$

  Solving the system of equations we find that

$$\begin{aligned} v^m(\varphi_H) &= \Big[\pi^m(\varphi_H) + \frac{\beta(1-\theta)}{1-\beta\theta}\pi^m(\varphi_L)\Big]\Big(\frac{1-\beta\theta}{1-2\beta\theta-\beta^2(1-2\theta)}\Big) \\ v^m(\varphi_L) &= \frac{\pi^m(\varphi_L) + \beta(1-\theta) v^m(\varphi_H)}{1-\beta\theta} \end{aligned}$$

  Note that $v^m(\varphi_H)$ is in $v^m(\varphi_L)$.

- The total output produced will be $Q^m(\varphi_i) = \varphi_i(n_i^*)^\alpha$.
- The only condition we need to check is that exit is not profitable in any state. It is sufficient to simply check $v^m(\varphi_L) \geq 0$.
- The following table provides the equilibrium values for this example. Note that aggregate output and prices are now state dependent.

Monopoly Example
$P^m(0.6) = 4.338$, $Q^m(0.6) = 0.662$
$P^m(1.0) = 3.750$, $Q^m(1.0) = 1.250$
$n^m(0.6) = 1.217$, $\pi^m(0.6) = -0.246, v^m(0.6) = 9.073$
$n^m(1.0) = 1.563$, $\pi^m(1.0) = 1.225, v^m(1.0) = 15.412$

5 Duopoly Example (NF=2)

In this case the price clears the market between firms playing a non-cooperative Cournot game. In particular, when there is more than one firm, each firm chooses the quantity of goods produced taking into account the production and exit strategies of other firms. In this case total output is given by $Q^d = \sum_{i=1}^{2} q^i$.

In this type of industry the state space is doubled. We need to consider all possible combinations of φ between the two possible firms induced by the idiosyncratic uncertainty. In fact the set of all possible states is given by $\{(\varphi_L, \varphi_L), (\varphi_L, \varphi_H), (\varphi_H, \varphi_L), (\varphi_H, \varphi_H), (\varphi_L, \varnothing), (\varphi_H, \varnothing), (\varnothing, \varnothing)\}$. Notice that in a symmetric duopoly equilibrium there are 3 possible aggregate states (i.e. boom, bust, and middle road). However, in certain states, one firm may choose to exit so that it adds more possible aggregate states (i.e. fluctuating between duopoly, monopoly, and possibly even autarky).

We will use the following notation. Let any function f (which could be a value function or decision rule or distribution of firms) be written $f(\varphi_s^i, \overline{\varphi}_{-i})$ where $\varphi_s^i \in \{\varnothing, \varphi_L, \varphi_H\}$ is the productivity shock of firm i (and if $\varphi_s^i = \varnothing$ it is understood that firm i is contemplating entry) and $\overline{\varphi}_{-i} \in \{\varnothing, \varphi_L, \varphi_H\}$ is the vector of all other possible productivity shocks $\overline{\varphi}_{-i}$ of potential competitors (with $NF = 2$, $\overline{\varphi}_{-i}$ is simply a scalar, not a vector). For example, $\pi^d(\varphi_H, \varphi_L)$ is the profit of a firm with own productivity φ_H whose competitor received φ_L. As another example, $\pi^d(\varphi_H, \varnothing)$ is the profit of the firm with own productivity φ_H currently in a monopoly state. Finally, $e^d(\varnothing, \varphi_s)$ denotes the entry decision of a potential firm when there is an incumbent firm with productivity φ_s.

The optimal labor choice remains static since there are no other endogenous state variables which enter the profit function than the distribution of incumbent firms. Hence,

- If only one firm produces the optimal labor choice is given by the solution to the monopolist problem. That is $\pi^d(\varphi_i, \varnothing) = \pi^m(\varphi_i)$. Note that the value of the other firm's productivity is irrelevant because it is not active.

- If the two firms produce, firm i solves

$$\pi^d(\varphi_i, \varphi_j) = \max_{n_i} \left[D - \varphi_i (n_i)^\alpha - \varphi_j (n_j)^\alpha \right] \varphi_i (n_i)^\alpha - n_i - c_f \tag{14}$$

  The FOC is

$$D\alpha\varphi_i (n_i^d)^{\alpha-1} - 2\varphi_i^2 \alpha \left(n_i^d\right)^{2\alpha-1} - \varphi_j n_j^\alpha \alpha \varphi_i \left(n_i^d\right)^{\alpha-1} = 1 \tag{15}$$

  This gives the labor demand policy function (given the second firm labor choice) of firm i, $n_i^d(\varphi_i, \varphi_j | n_j)$.

- Specifically, suppose $\alpha = 1/2$. Then (15) can be written

$$n_i^d = \left(\frac{\varphi_i \left[D - \varphi_j n_j^{1/2} \right]}{2 \left[1 + \varphi_i^2 \right]} \right)^2$$

- Note that in the foc (15), firm i takes as given its competitor's actions n_j. A Cournot-Nash equilibrium requires consistency; that is, firm $i's$ belief about n_j in (15) is consistent with optimal actions taken by firm j. Specifically, in equilibrium $n_j = n_j^d(\varphi_j, \varphi_i | n_i)$.

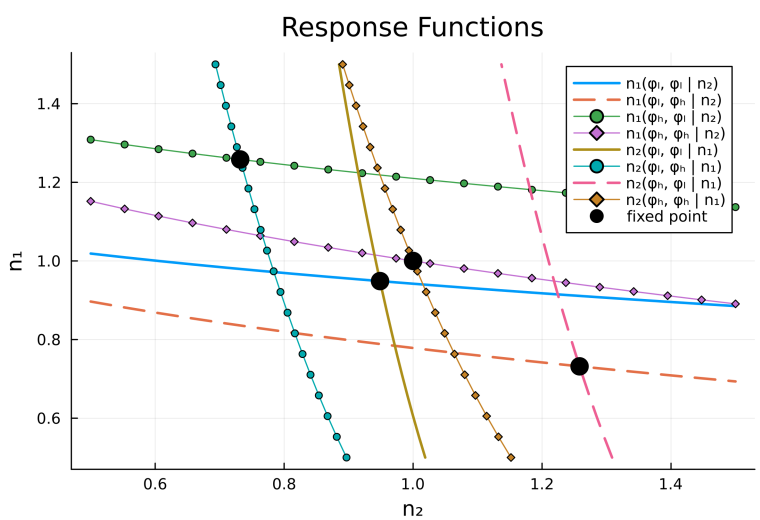

Figure 5: Duopoly Response Functions

- Conditional on holding firm j's output constant (i.e. n_j and φ_j), labor at firm i rises if productivity there rises (of course with a finite set of φ we can't literally take derivatives, but let's be a bit sloppy here):

$$\begin{aligned}\frac{dn_i}{d\varphi_i} &= 2\left(\frac{\varphi_i\left[D-\varphi_j n_j^{1/2}\right]}{2\left[1+\varphi_i^2\right]}\right)\cdot\left[\frac{2\left[1+\varphi_i^2\right]\left[D-\varphi_j n_j^{1/2}\right]-\varphi_i\left[D-\varphi_j n_j^{1/2}\right]4\varphi_i}{4\left[1+\varphi_i^2\right]^2}\right]\\ &= 2\left(\frac{\varphi_i\left[D-\varphi_j n_j^{1/2}\right]}{2\left[1+\varphi_i^2\right]}\right)\cdot\left[\frac{\left[D-\varphi_j n_j^{1/2}\right]\left[1-\varphi_i^2\right]}{2\left[1+\varphi_i^2\right]^2}\right]>0.\end{aligned}$$

- The decision rules $n_i^d(\varphi_i,\varphi_j|n_j^d)$ and $n_j^d(\varphi_j,\varphi_i|n_i^d)$ can also be thought of as reaction functions. The slope of the reaction function of firm i is given by

$$\frac{dn_i^d}{dn_j}=-\left(\frac{\varphi_i\left[D-\varphi_j n_j^{1/2}\right]}{2\left[1+\varphi_i^2\right]}\right)\frac{\varphi_i\left[\varphi_j n_j^{-1/2}\right]}{2\left[1+\varphi_i^2\right]}<0.$$

- We plot reaction functions in Figure 1 for a particular equilibrium where firms with a low productivity shock exit (hence any reaction function that has a vector where $\varphi_i \neq \varphi_j$ is not consistent with behavior on-the-equilibrium path, but of course we still need to solve for best responses off-the-equilibrium path). As can be seen in the figure, the higher the common shocks on-the-equilibrium path, the higher demand for labor. Also, in the asymmetric off-the-equilibrium path case where there are two incumbents with $\varphi_i \neq \varphi_j$,the figure shows that the firm with the high productivity shock demands more labor than the symmetric case.

- If two firms are in the market there are four possible states of the world (φ_L,φ_L), (φ_L,φ_H), (φ_H,φ_L) and (φ_H,φ_H).

- The value function is given by

$$v(\varphi_i,\varphi_{-i})=\max\Big\{0,\pi(\varphi_i,\varphi_{-i})+\beta E\big[v(\varphi_i',\varphi_{-i}')\big]\Big\}.$$

- As before we can write this expression as:

$$v(\varphi_i,\varphi_{-i})=\max_{x\in\{0,1\}}(1-x)\Big\{\pi(\varphi_i,\varphi_{-i})+\beta E\big[v(\varphi_i',\varphi_{-i}')\big]\Big\}$$

  which yields exit decision rules $x(\varphi_i,\varphi_{-i})$.

5.1 An Equilibrium where NF=2 binds

- We will construct an equilibrium with the following characteristics:
  - If two firms are active (i.e. $NF=2$), Table 2

(φ_i, φ_j)	Exit Decision
(φ_L, φ_L)	$(stay, stay)$
(φ_L, φ_H)	$(exit, stay)$
(φ_H, φ_L)	$(stay, exit)$
(φ_H, φ_H)	$(stay, stay)$

Table 2:

  - If only one firm is active it always stays in the market (i.e. $x(\varphi_i, \varnothing) = 0$ for all $\varphi_i \in \{\varphi_L, \varphi_H\}$).
  - As for entry, a firm enters only if a monopolist receives φ_L. Since $NF = 2$, it would be infinitely costly to enter with two active incumbents. If there are no incumbents (which never happens on-the-equilibrium-path of this construction but could happen off the equilibrium path), then two potential entrants are randomly selected to enter.

- Note that in this environment the economy will fluctuate between duopoly and monopoly of random lengths. For instance, if there is a realization of shocks such that there is asymmety in the duopoly case, then we enter a monopoly phase until the monopolist recieves a low shock then a duopoly will arise in the following period.

- The 6 value functions become

$$
\begin{aligned}
v(\varphi_L, \varphi_L) &= \pi(\varphi_L, \varphi_L) + \beta\Big\{\theta\big[\theta v(\varphi_L, \varphi_L) + (1-\theta)\cdot 0\big] + \\
&\quad (1-\theta)\big[\theta v(\varphi_H, \varphi_L) + (1-\theta) v(\varphi_H, \varphi_H)\big]\Big\} \\
v(\varphi_L, \varphi_H) &= 0 \\
v(\varphi_H, \varphi_L) &= \pi(\varphi_H, \varnothing) + \beta\Big\{\theta v(\varphi_H, \varnothing) + (1-\theta) v(\varphi_L, \varnothing)\Big\} \\
v(\varphi_H, \varphi_H) &= \pi(\varphi_H, \varphi_H) + \beta\Big\{(1-\theta)\big[\theta\cdot 0 + (1-\theta) v(\varphi_L, \varphi_L)\big] + \\
&\quad \theta\big[\theta v(\varphi_H, \varphi_H) + (1-\theta) v(\varphi_H, \varphi_L)\big]\Big\} \\
v(\varphi_H, \varnothing) &= \pi(\varphi_H, \varnothing) + \beta\Big\{\theta v(\varphi_H, \varnothing) + (1-\theta) v(\varphi_L, \varnothing)\Big\} \\
v(\varphi_L, \varnothing) &= \pi(\varphi_L, \varnothing) + \beta\Big\{\theta\big[(1-\eta) v(\varphi_L, \varphi_e = \varphi_L) + \eta\cdot 0\big] + \\
&\quad (1-\theta)\big[(1-\eta) v(\varphi_H, \varphi_e = \varphi_L) + \eta v(\varphi_H, \varphi_e = \varphi_H)\big]\Big\}
\end{aligned}
$$

This is a system of 6 equations in 6 unknowns that you can solve with the help of the computer.

- From the solution to this system of equations, you can pin down the highest and the lowest value of the entry cost, $(\underline{c}, \overline{c})$, such that this equilibrium exists. That is, if $N = 1$, it must be profitable to enter if the incumbent firm currently has φ_L and not to enter if the incumbent firm has φ_H. To check this (recalling that entry comes one period in advance (which explains β)):

  - If incumbent has φ_L, expected value of entry $\geq c_e$ or

$$\beta \left\{ \begin{array}{c} (1-\eta)\left[(\theta v(\varphi_e = \varphi_L, \varphi_L) + (1-\theta)\cdot 0\right] \\ +\eta\left[\theta v(\varphi_e = \varphi_H, \varphi_L) + (1-\theta) v(\varphi_e = \varphi_H, \varphi_H)\right] \end{array} \right\} \geq c_e. \quad (16)$$

    Let $\overline{c}$ denote the value of the left hand side of (16). Then entry occurs provided costs are below that number (i.e. $\overline{c} \geq c_e$).

  - If incumbent has φ_H, expected value of entry $< c_e$ or

$$\beta \left\{ \begin{array}{c} (1-\eta)\left[(\theta \cdot 0 + (1-\theta) v(\varphi_e = \varphi_L, \varphi_L)\right] \\ +\eta\left[\theta v(\varphi_e = \varphi_H, \varphi_H) + (1-\theta) v(\varphi_e = \varphi_H, \varphi_L)\right] \end{array} \right\} < c_e \quad (17)$$

    Let $\underline{c}$ denote the value of the left hand side of (17). Then entry does not occur if costs are above that number (i.e. $\underline{c} < c_e$).

  - Notice that if $v(\varphi_L, \varphi_L) < v(\varphi_H, \varphi_H) < v(\varphi_H, \varphi_L)$, then it is possible that the lhs of (16) exceeds the lhs of (17) since $\theta > (1-\theta)$.

- Hence costs which satisfy the condition for entry when the incumbent has φ_L and no entry when the incumbent has φ_H must satisfy $c_e \in (\underline{c}, \overline{c}]$. Generally we should prove that this set is non-empty. In the particular parameterization above, $\underline{c} = 2.36$ and $\overline{c} = 4.90$.

- Also, we need to verify that in the off-the-equilibrium path event that there are no incumbents at all, two randomly selected entrants would choose to enter at $c_e \in (\underline{c}, \overline{c}]$. That is

$$\beta \left\{ (1-\eta)^2 v(\varphi_L, \varphi_L) + (1-\eta)\eta v(\varphi_L, \varphi_H) + \eta(1-\eta) v(\varphi_H, \varphi_L) + \eta^2 v(\varphi_H, \varphi_H) \right\} \geq c_e. \quad (18)$$

  In this particular case, the left hand side equals 3.63, so there will be entry by two firms.

- Next, we must check that no firm has an incentive to deviate (for one period) from what we conjectured in the above table.

  - Row 1: $v(\varphi_L, \varphi_L) \geq 0$.
  - Row 2: This is the most important condition - we need to check that exit is profitable for an incumbent firm with φ_L when the other incumbent firm has φ_H, i.e. $v(\varphi_L, \varphi_H) = 0$. Hence, we must check that the incumbent firm with φ_L does not deviate from exiting in this state. Then since the value from such a deviation is given by

$$v^{dev}(\varphi_L, \varphi_H) = \pi(\varphi_L, \varphi_H) + \beta \left\{ \begin{array}{c} \theta\left[\theta \cdot 0 + (1-\theta) v(\varphi_L, \varphi_L)\right] \\ +(1-\theta)\left[\theta v(\varphi_H, \varphi_H) + (1-\theta) v(\varphi_H, \varphi_L)\right] \end{array} \right\}$$

we need $v^{dev}(\varphi_L, \varphi_H) < 0 = v(\varphi_L, \varphi_H)$. Note the 0 in v^{dev} occurs since the firm returns to being on the equilibrium path in state (φ_L, φ_H) and that $v(\varphi_H, \varphi_L)$ ensures the deviating firm enjoys monopoly profits in that period.

  - Row 3: $v(\varphi_H, \varphi_L) \geq 0$.
  - Row 4: $v(\varphi_H, \varphi_H) \geq 0$

- Finally, to check that a monopolist does not deviate from what we conjectured above and exit, we need:

$$\begin{aligned} v(\varphi_H, \varnothing) &\geq 0 \\ v(\varphi_L, \varnothing) &\geq 0. \end{aligned}$$

- The following table provides the equilibrium values for this example. Note that aggregate output and prices are now state dependent.

Duopoly Example
$P^d(0.6, 0.6) = 3.831$, $Q^d(0.6) = 1.169$
$P^d(1.0, 1.0) = 3.000$, $Q^d(1.0, 1.0) = 2.000$
$n^d(0.6, 0.6) = 0.949$, $\pi^d(0.6, 0.6) = -0.610, v^d(0.6, 0.6) = 1.200$
$n^d(1.0, 1.0) = 1.0$, $\pi^d(1.0, 1.0) = 0.100, v^d(1.0, 1.0) = 4.259$

- Comparing all cases we have:

	Competitive	Monopoly	Duopoly
Prices	$P = 3.296$	$P^m(0.6) = 4.338$	$P^d(0.6, 0.6) = 3.831$
		$P^m(1.0) = 3.750$	$P^d(1.0, 1.0) = 3.000$
Quantities	$Q = 1.704$	$Q^m(0.6) = 0.662$	$Q^d(0.6) = 1.169$
		$Q^m(1.0) = 1.250$	$Q^d(1.0, 1.0) = 2.000$
Profits	$\pi(0.6) = -0.922$(o-e-p)	$\pi^m(0.6) = -0.246$	$\pi^d(0.6, 0.6) = -0.610$
	$\pi(1.0) = 0.816$	$\pi^m(1.0) = 1.225$	$\pi^d(1.0, 1.0) = 0.100$
Labor Demand	$n(0.6) = 0.994$	$n^m(0.6) = 1.217$	$n^d(0.6, 0.6) = 0.949$
	$n(1.0) = 2.716$	$n^m(1.0) = 1.563$	$n^d(1.0, 1.0) = 1.0$
Value of firm	$v(0.6) = 0$ (o-e-p)	$v^m(0.6) = 9.073$	$v^d(0.6, 0.6) = 2.666$
	$v(1.0) = 6.0$	$v^m(1.0) = 15.412$	$v^d(1.0, 1.0) = 5.725$

- One can then check whether the model generates procylical entry, countercylical exit, and countercylical markups (see Jaimovich and Floetotto (2008, Journal of Monetary Economics) for a DSGE model with imperfect competition with these properties).

  - Obviously, the competitive case cannot generate this unless we add aggregate uncertainty.
  - The monopoly case does not have entry and exit, but the markup $P^m(\varphi) - 1$ is countercyclical.
  - The duopoly case also has countercyclical markups, countercyclical exit, but predicts countercylical entry.

5.2 An Equilibrium where NF=2 doesn't bind

Here we check whether it is possible to find a monopoly equilibrium even though the costs are not infinite to enter if there is already an incumbent. We will show that there is an equilibrium with only one firm operating (a monopolist) under this structure and he always stays. This means that $NF = 2$ does not constrain entry. Basically, we simply need that c_e is high enough to deter entry if there is already an incumbent.

- We construct an equilibrium almost identical to what we had before except that if there is one firm, there is no entry for all $\varphi_i \in \{\varphi_L, \varphi_H\}$ of the incumbent. That is, we keep table 2 as a conjecture for the off-the-equilibrium path node where two firms are active.
- The 6 value functions become

$$\begin{aligned}
v(\varphi_L,\varphi_L) &= \pi(\varphi_L,\varphi_L) + \beta\Big\{\theta\big[\theta v(\varphi_L,\varphi_L) + (1-\theta)\cdot 0\big] + \\
&\quad (1-\theta)\big[\theta v(\varphi_H,\varphi_L) + (1-\theta)v(\varphi_H,\varphi_H)\big]\Big\} \\
v(\varphi_L,\varphi_H) &= 0 \\
v(\varphi_H,\varphi_L) &= \pi(\varphi_H,\varnothing) + \beta\Big\{\theta v(\varphi_H,\varnothing) + (1-\theta)v(\varphi_L,\varnothing)\Big\} \\
v(\varphi_H,\varphi_H) &= \pi(\varphi_H,\varphi_H) + \beta\Big\{(1-\theta)\big[\theta\cdot 0 + (1-\theta)v(\varphi_L,\varphi_L)\big] \\
&\quad \theta\big[\theta v(\varphi_H,\varphi_H) + (1-\theta)v(\varphi_H,\varphi_L)\big]\Big\} \\
v(\varphi_H,\varnothing) &= \pi(\varphi_H,\varnothing) + \beta\Big\{\theta v(\varphi_H,\varnothing) + (1-\theta)v(\varphi_L,\varnothing)\Big\} \\
v(\varphi_L,\varnothing) &= \pi(\varphi_L,\varnothing) + \beta\Big\{\theta v(\varphi_L,\varnothing) + (1-\theta)v(\varphi_H,\varnothing)\Big\}
\end{aligned}$$

  Note the only difference from the previous one is the last line.

- Next, we must check that no firm has an incentive to deviate from what we conjectured in the above table.
  - Row 1: $v(\varphi_L,\varphi_L) \geq 0$.
  - Row 2: Since the value from such a deviation is given by

$$\begin{aligned}
v^d(\varphi_L,\varphi_H) &= \pi(\varphi_L,\varphi_H) + \beta\Big\{\theta\left[\theta\cdot 0 + (1-\theta)v(\varphi_L,\varphi_L)\right] \\
&\quad +(1-\theta)\left[\theta v(\varphi_H,\varphi_H) + (1-\theta)v(\varphi_H,\varphi_L)\right]\Big\}
\end{aligned}$$

    we need $v^d(\varphi_L,\varphi_H) < 0 = v(\varphi_L,\varphi_H)$.
  - Row 3: $v(\varphi_H,\varphi_L) \geq 0$.
  - Row 4: $v(\varphi_H,\varphi_H) \geq 0$

- To check that a monopolist does not deviate from what we conjectured above and exit, we need:

$$\begin{aligned} v(\varphi_H, \varnothing) &\geq 0 \\ v(\varphi_L, \varnothing) &\geq 0. \end{aligned}$$

- Also, we must check that no one enters to compete with the monopolist with φ_L (it is sufficient to check when the monopolist has a low shock, since if this condition holds, then a potential entrant would surely not enter if the monopolist has the high shock.):

$$\beta \left\{ \begin{array}{c} (1-\eta)\left[(\theta v(\varphi_e = \varphi_L, \varphi_L) + (1-\theta)\cdot 0\right] \\ +\eta\left[\theta v(\varphi_e = \varphi_H, \varphi_L) + (1-\theta) v(\varphi_e = \varphi_H, \varphi_H)\right] \end{array} \right\} < c_e.$$

If you kept all the parameters the same, except let $c_e = 5$ instead of $c_e = 3$, then you would get this type of equilibrium since the value on the lhs is 4.91.